Curve Bio – AI Research Intern (Genomic Foundation Models)

Company: [Curve Bio]
Location: San Mateo, CA
Type: Hybrid, remote if needed
Description:

Curve Bio is building AI-driven, biologically rooted insights to transform chronic disease care. By combining large-scale next-generation sequencing (NGS) data with modern machine learning, we aim to develop foundation models that capture biologically meaningful signals beyond raw sequence.

We are seeking a highly motivated undergraduate intern to join our Computational Sciences team for a 10-week research-focused internship. In this role, you will contribute to advancing AI models for genomics, with a focus on understanding how large-scale sequence models adapt to new biological modalities and disease-relevant signals.

You will work closely with computational scientists to design experiments, evaluate model behavior, and translate representation learning advances into biologically meaningful insights.

Project Overview:

Modern transformer-based models have shown strong performance on DNA sequence modeling tasks. However, biological regulation involves multiple layers of information beyond primary sequence.

This internship will focus on:

- Evaluating how genomic foundation models respond to new biological signal types
- Designing benchmarking tasks to quantify improvements in disease-relevant prediction
- Performing controlled experiments to understand model adaptation and generalization
- Conducting systematic ablation and robustness studies
- Analyzing internal model representations to better interpret learned biological structure

The project sits at the intersection of representation learning, genomics, and disease modeling, and aims to strengthen the scientific rigor and translational potential of foundation models in biology.

Desired Skills:

- Currently pursuing a Bachelor's degree in Computational Biology, Computer Science, Data Science, Mathematics, Statistics, Bioengineering, or a related field
- Strong programming skills in Python
- Familiarity with PyTorch and/or transformer-based models
- Working knowledge of machine learning fundamentals (supervised learning, representation learning, model evaluation)
- Experience with data analysis in the SciPy/PyData ecosystem (numpy, pandas, matplotlib, etc.)
- Interest in genomics, epigenetics, or next-generation sequencing data
- Comfort working in Linux environments and using Git for version control

- Strong analytical thinking and ability to translate biological questions into computational experiments
- Excellent written and verbal communication skills

Bonus qualifications (not required):
- Experience with large language models (LLMs) or fine-tuning transformer architectures
- Familiarity with NGS data formats (e.g., FASTQ, BAM, BED)
- Prior research experience in computational biology or applied machine learning

Expected Outcomes:

By the end of the internship, you will:
- Design and execute rigorous computational experiments
- Build reproducible evaluation pipelines
- Produce quantitative benchmarks demonstrating model performance and robustness
- Deliver a final technical presentation summarizing findings and future directions
- Contribute to internal research reports and potentially publication-ready analyses